Muneeb Lone
23i-2623
DS-B

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Homework #8

Q1: $M = \begin{bmatrix} d-z_c & 0 & -x_c & x_c d \\ 0 & d-z_c & -y_c & y_c d \\ 0 & 0 & d & -d z_c \\ 0 & 0 & 1 & -z_c \end{bmatrix} = \begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

$A(0,0,0)$: $Img(A) = MA = \begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ -2 \\ 1 \end{bmatrix}$

$B(1,0,0)$: $Img(B) = MB = \begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ -2 \\ 1 \end{bmatrix}$

$C(0,1,0)$: $Img(C) = MC = \begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \\ -2 \\ 1 \end{bmatrix}$

$D(1,1,1)$: $Img(D) = MD = \begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 0 \\ 2 \end{bmatrix}$

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Q2 (i): $T_E \begin{bmatrix} 2a-b \\ a+b-3c \\ c-a \end{bmatrix}$ Marks 2

$b = 2a$

$a = c$

$T = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 1 & -3 \\ -1 & 0 & 1 \end{bmatrix}$ $\ker(T) = \text{Span}\{1+2x+x^2\}$

$\ker(T) = \left[\begin{array}{ccc|c} 2 & -1 & 0 & 0 \\ 1 & 1 & -3 & 0 \\ -1 & 0 & 1 & 0 \end{array}\right]$

$\left[\begin{array}{ccc|c} \textcircled{1} & 1 & -3 & 0 \\ 2 & -1 & 0 & 0 \\ -1 & 0 & 1 & 0 \end{array}\right]$ $R2 \leftrightarrow R1$

$\left[\begin{array}{ccc|c} \textcircled{1} & 1 & -3 & 0 \\ 0 & -3 & 6 & 0 \\ 0 & 1 & -2 & 0 \end{array}\right]$ $R2-2R1$ $R3+R1$

$\left[\begin{array}{ccc|c} \textcircled{1} & 1 & -3 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 1 & -2 & 0 \end{array}\right]$ $-R2/3$

$\left[\begin{array}{ccc|c} \textcircled{1} & 1 & -3 & 0 \\ 0 & \textcircled{1} & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ $R3-R2$

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- Ker (T) $\neq \{0\}$ so not one to one
- Columns linearly dependent so does not span entire space hence not onto.
- Not isomorphic.

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(ii) $T: M_{22} \to M_{22}$ by $T(A) = AB$, where $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

$E_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $E_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $E_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $E_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

$$T(E_1) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$$T(E_2) = \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}, \quad T(E_3) = \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix}, \quad T(E_4) = \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix}$$

$\ker(T)$ will consist of matrices that lie in the nullspace of the transformation since B has rank 1 and is non-invertible. $\ker(T) \neq 0$ hence not one to one.

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$$\ker(T) = A\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$ Multiply

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} a-b & -a+b \\ c-d & -c+d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \therefore a=b,\ c=d$$

$$A = \begin{bmatrix} a & a \\ c & c \end{bmatrix}$$

$$\ker(T) = \left\{ \begin{bmatrix} a & a \\ c & c \end{bmatrix} \in \mathbb{R} \mid a, c \in \mathbb{R} \right\}$$

$$\ker(T) = \text{Span}\left(\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \right)$$

~~$\text{Im}(T)(A) = T(A)$~~ $\begin{bmatrix} a-b & -a+b \\ c-d & -c+d \end{bmatrix} = \begin{bmatrix} x & -x \\ y & -y \end{bmatrix}$

$$\text{Im}(T) = \left\{ \begin{bmatrix} x & -x \\ y & -y \end{bmatrix} \mid x, y \in \mathbb{R} \right\} = \text{Span}\left(\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} \right)$$

T is not one to one ($\ker(T) \neq \{0\}$) nor on to ($\text{Im}(T) \in M_{22}$) and is hence not an isomorphism.

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(iii) $T: \mathbb{R}^3 \to W$ by $T\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a+b+c & b-2c \\ b-2c & a-c \end{bmatrix}$ Linear

W is vector space of all symmetric 2×2 matrices.

$$T\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a+b+c & b-2c \\ b-2c & a-c \end{bmatrix} \Rightarrow$$

$$\ker(T) = \begin{bmatrix} a+b+c & b-2c \\ b-2c & a-c \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\left.\begin{aligned} a+b+c &= 0 \\ b-2c &= 0 \\ a-c &= 0 \end{aligned}\right\} \quad 4c = 0 \to c = 0$$

As $a = b = c$, $\ker(T) = \{0\}$

~~a+b=c~~

$$a\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{X} + b\underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}}_{Y} + c\underbrace{\begin{bmatrix} 1 & -2 \\ -2 & -1 \end{bmatrix}}_{Z}$$

Since X, Y and Z are linearly independent.

$$\therefore Im(T) = W$$

As $\ker(T) = \{0\}$ and $Im(T) = W$, T is isomorphic.

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Q3: $T: \mathbb{R}^2 \to \mathbb{P}_2$ $\quad T\begin{bmatrix}1\\1\end{bmatrix} = 1-2x$, $T\begin{bmatrix}3\\-1\end{bmatrix} = x+x^2$

$$T\begin{bmatrix}-7\\9\end{bmatrix} = \begin{bmatrix}a\\b\end{bmatrix} = \alpha\begin{bmatrix}1\\1\end{bmatrix} + \beta\begin{bmatrix}3\\-1\end{bmatrix}$$

Hint

$$\left[\begin{array}{cc|c} 1 & 3 & a \\ 1 & -1 & b \end{array}\right] \qquad \left[\begin{array}{cc|c} 1 & 3 & a \\ 0 & 1 & (-b+a)/4 \end{array}\right] -R2/4$$

$$\sim \left[\begin{array}{cc|c} 1 & 3 & a \\ 0 & -4 & b-a \end{array}\right] R2-R1 \qquad \left[\begin{array}{cc|c} 1 & 0 & a-\left(\frac{3}{4}\right)(-b+a) \\ 0 & 1 & (-b+a)/4 \end{array}\right] R1-3R2$$

$-4\beta = b-a$ $\qquad \alpha + 3\beta = a$

$\alpha = a - \frac{3}{4}(-b+a)$

$-4\beta = b-a-3\beta$

$\beta = \frac{-b+a}{4}$

$\beta = b-\alpha$

$$T\begin{bmatrix}a\\b\end{bmatrix} \quad \alpha = \frac{a+3b}{4}, \quad \beta = \frac{a-b}{4}$$

(i) $$T\begin{bmatrix}-7\\9\end{bmatrix} = \alpha T\begin{bmatrix}1\\1\end{bmatrix} + \beta T\begin{bmatrix}3\\-1\end{bmatrix}$$

$$= \left(\frac{-7+3(9)}{4}\right)(1-2x) + \left(\frac{-7-9}{4}\right)(x+x^2)$$

$$= (5)(1-2x) + (-4)(x+x^2)$$

$$= 5-14x-4x^2 \quad \leftarrow \text{Ans}$$

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(ii) General form for $T\begin{bmatrix} a \\ b \end{bmatrix}$:

$$T\begin{bmatrix} a \\ b \end{bmatrix} = \frac{a+3b}{4} - \frac{(a+3b)}{4}x + \frac{(a-b)}{4}x + \frac{(a-b)}{4}x^2$$